

CUSTOMER BEHAVIOR ANALYSIS

Objective

The objective of this project is to analyze customer transactions and behavior to identify purchase patterns, customer segments, and churn risks. The analysis helps in understanding customer preferences and provides actionable recommendations to improve engagement and retention for Alfido Tech.

Dataset Description

The dataset contains customer transaction records with information such as:

- Customer ID
- Purchase Date
- Product Category
- Product Price
- Quantity
- Total Purchase Amount
- Payment Method
- Customer Age
- Gender
- Returns
- Churn

The dataset consists of approximately 250,000 records and 13 columns.

Data Cleaning

The following preprocessing steps were performed:

- Loaded the dataset using Pandas.
 - Examined data types and summary statistics.
 - Checked for missing values.
 - Removed duplicate records.
 - Converted the Purchase Date column to datetime format.
 - Prepared the data for analysis and customer segmentation.
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Exploratory Data Analysis

Summary statistics including mean, median, and mode were calculated for numerical variables. Multiple visualizations were created to understand customer behavior and transaction patterns.

The following analyses were performed:

- Product category distribution
 - Gender distribution
 - Customer churn analysis
 - Payment method preferences
 - Purchase patterns
 - Top spending customers
 - Customer segmentation
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Customer Segmentation using RFM Analysis

RFM (Recency, Frequency, Monetary) analysis was used to segment customers based on:

Recency

Measures how recently a customer made a purchase.

Frequency

Measures how often a customer makes purchases.

Monetary

Measures the total amount spent by a customer.

Using these metrics, customers were divided into different segments to identify high-value and low-value groups.

Purchase Patterns

Analysis of product categories revealed varying customer preferences across different categories such as Electronics, Books, Home, Clothing, and Sports.

Certain categories recorded higher purchase frequencies, indicating stronger customer demand.

Retention Trends

Customer churn analysis showed that most customers remain active, while a smaller group exhibits a higher risk of leaving. Understanding churn patterns can help improve customer retention strategies.

Key Findings

1. Customer purchasing behavior varies significantly across product categories.
 2. High-value customers contribute a substantial portion of total revenue.
 3. Different payment methods are preferred by customers.
 4. RFM analysis successfully identified multiple customer segments.
 5. Churn analysis highlighted the importance of customer retention strategies.
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Recommendations for Alfido Tech

1. Implement Loyalty Programs

Reward high-value customers through discounts and exclusive offers.

2. Personalized Marketing

Provide recommendations based on customers' previous purchase behavior.

3. Reduce Customer Churn

Target customers at risk with promotional campaigns and personalized communication.

4. Focus on Popular Product Categories

Allocate resources and marketing efforts toward high-performing categories.

5. Enhance Customer Experience

Optimize payment options and improve the checkout process to increase customer satisfaction.

Conclusion

This analysis provided valuable insights into customer behavior, spending patterns, and retention trends. Customer segmentation through RFM analysis enabled the identification of valuable customer groups and churn risks. The findings and recommendations can help Alfido Tech improve customer engagement, retention, and overall business performance.